MIL-E-15330D(SH)
14 October 1977
SUPERSEDING
MIL-E-15330C(SHIPS)
10 August 1965, and
MIL-E-23616(SHIPS)
24 May 1965
(See 6.5)

MILITARY SPECIFICATION

EXPANSION JOINT, PIPE, NONMETALLIC, FIRE-RETARDANT

1. SCOPE

1.1 Scope. This specification covers the construction and material requirements for fire-retardant, nonmetallic expansion joints.

1.2 Classification. Expansion joints shall be of the following classes and types, and shall be similar to those shown on figures 1 through 6, as specified (see 6.2.1).

Class A - Synthetic rubber, flanged.

Type I - Single arch (sea water service).
Type II - Rectangular type (vacuum service).
Type III - U-type (vacuum service).

Class B - Synthetic rubber, non-flanged, corrugated.
Class C - TFE fluorocarbon resin, corrugated, with ductile iron flanges.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
QQ-S-766 - Steel Plates, Sheets, and Strips Corrosion-Resisting.
QQ-W-423 - Wire, Steel, Corrosion-Resisting.
PPP-B-601 - Box, Wood, Cleated, Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated-Triple-Wall.
PPP-C-650 - Crates, Wood, Open and Covered.

MILITARY

MIL-P-116 - Preservation-Packaging Methods of.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-R-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-B-13239 - Barrier Material, Waterproofed, Flexible, All Temperature, Heat Sealable.
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-C-20079 - Cloth, Glass; Tape, Textile Glass; and Thread, Glass.
MIL-R-20092 - Rubber Sheets and Molded Shapes, Cellular, Synthetic Open Cell (Foam Latex).
MIL-C-20696 - Cloth, Coated, Nylon, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20632 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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MILITARY (Continued)

MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion-Resisting, Austentic, Low Magnetic Permeability) (For Shipboard Application).

STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-271 - Nondestructive Testing Requirements For Metals.
MIL-STD-278 - Fabrication Welding and Inspection; And Casting Inspection and Repair For Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-407 - Visual Guide for Rubber Molded Stems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B16.5 - Steel Pipe Flanges, Flanged Valves and Fittings.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 638 - Tensile Properties of Plastics, Test for.
D 1457 - TFE Fluorocarbon Resin Molding and Extrusion Material.
D 1708 - Tensile Properties of Plastics by Use of Microtensile Specimens, Test for.
E 8 - Tension Testing of Metallic Materials.

(Application for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Classes and Rules

(Application for copies should be addressed to the ATA Tariff Section, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings Rules, and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, G.F. Earl, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be examined and tested as specified in 4.2 (see 6.3).

3.2 Construction and materials.

3.2.1 Class A expansion joints. Class A expansion joints (see figures 1 through 3) shall be fabricated of a synthetic rubber tube covered with multiple plies of woven cloth covered with a synthetic rubber cover. Inner tube and outer cover shall be a copolymer of butadiene and acrylonitrile. The cover stock shall be compounded for oil resistance and

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fire retardance. The reinforcing cloth shall be thoroughly impregnated with a compatible friction stock. Additional reinforcement of the class A, type I expansion joint may be provided by solid or spiral reinforcing rings or by a flat braided wire ribbon imbedded in a synthetic rubber filler compatible with the basic construction elastomer. Metal parts shall be treated by phosphatizing, galvanizing, or copper coating prior to being imbedded in the expansion joint carcass. The treated reinforcement wire shall be in accordance with table I and shall withstand the tests specified in 4.6.1. Reinforcing rings shall be welded prior to installation on expansion joint body. Welds shall be 100 percent penetration.

TABLE I. Tensile properties.

Wire diameter	Tensile strength	
Inch	Minimum lb/in^2	Maximum lb/in^2
0.035 to 0.105, inclusive	180,000	300,000
Larger than 0.105 to 0.207, inclusive	150,000	240,000
Larger than 0.207 to 0.362, inclusive	120,000	210,000

3.2.1.1 Reinforcing cloth. Reinforcing cloth shall be woven of glass, polyester or polyamide fibers, or a combination thereof. All reinforcing cloth shall be thoroughly impregnated with synthetic rubber. Fiberglass shall be impregnated resorcinol formaldehyde latex (R.F.L.) having adhesion characteristics that are compatible with natural and nytrile rubber, neoprene, and SBR.

3.2.1.2 Flanges. Flanges shall be integral with the body of the expansion joint. The tube, fabric plies and cover shall be brought up into the flange. Additional impregnated fabric plies or wire reinforcement may be included between the bolt holes as necessary to meet service conditions. Flanges of class A, type I and type III shall be drilled in accordance with MIL-F-20042 or ANSI B16.5 as specified (see 6.2.1). Flange drilling of class A, type II shall be as specified (see 6.2.1). Reinforcing fabric which is exposed by drilling of bolt holes shall be thoroughly coated with a fire retardant, self curing synthetic rubber.

3.2.1.3 Arch construction. Unless otherwise specified (see 6.2.1), the arch construction of the class A type I expansion joint shall be either a sharp arch in accordance with figure 1 or a long radius arch.

3.2.1.4 Retaining rings. Retaining rings shall be hot-rolled low carbon steel galvanized by hot dip process to prevent rusting. The rings shall be of a split-ring type, and shall be of sufficient width to extend from the body of the expansion joint to the outer edge of the flange. The edge adjacent to the body shall be beveled 1/8 inch and cut to sufficient size to prevent any cutting or bruising of the flange or body. The rings shall be sized for the intended diameter and working pressure with which the joint is to be used. The two ring sections provided for each flange shall be drilled in accordance with the flange drilling, and shall be split at bolt holes to insure a proper seal at all points. Expansion joints shall be shipped with retaining rings securely attached thereto.

3.2.1.5 Physical and mechanical properties. Physical and mechanical properties of class A, type I expansion joint shall be in accordance with table II for new construction. Replacement expansion joints not in dimensional accordance with table II shall be in accordance with table III and as specified (see 6.2.1). Class A, type II expansion joints shall be as specified (see 6.2.1).

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TABLE II. Physical and mechanical properties, class A, type I.

Pipe size (inches)	Face-to-face inches ± 1/8	Design Pressure lb/in^2	Proof Pressure lb/in^2	Design movement		
				Compression inch	Elongation inch	Lateral inch
2	6	150	225	7/16	1/4	1/2
2-1/2	6	150	225	7/16	1/4	1/2
3	6	150	225	7/16	1/4	1/2
4	6	150	225	7/16	1/4	1/2
5	6	125	190	7/16	1/4	1/2
6	6	125	190	7/16	1/4	1/2
8	6	125	190	11/16	3/8	1/2
10	8	125	190	11/16	3/8	1/2
12	8	125	190	11/16	3/8	1/2
14	8	75	110	11/16	3/8	1/2
16	8	60	90	11/16	3/8	1/2
18	8	60	90	11/16	3/8	1/2
20	8	60	90	13/16	7/16	1/2
24	10	60	90	13/16	7/16	1/2
28	10	50	75	15/16	1/2	1/2
30	10	50	75	15/16	1/2	1/2
36	10	50	75	15/16	1/2	1/2
38	10	50	75	15/16	1/2	1/2
42	12	50	75	1	9/16	1/2
48	12	50	75	1	9/16	1/2
46	12	50	75	1	9/16	1/2

TABLE III. Dimensional properties of non-standard expansion joints class A, type I (for replacement of existing expansion joints only - not for new construction).

Pipe size	Face-to-face (± 1/8 inch)
Inches	Inches
25	10
26	8-1/2
32	12
33	24
28	13
30	13
34	12
35	12
36	12
28	12
38	15
42	13

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3.2.1.6 Physical properties of synthetic rubber (class A expansion joint). The physical properties of the synthetic rubber for class A expansion joints shall be in accordance with table IV when tested in accordance with 4.6.2.

TABLE IV. Physical properties.

	Initial	After aging
Tensile, lb/in^2 minimum	1200	960
Elongation, percent minimum	500	325

3.2.2 Class B expansion joint. The body of class B expansion joints (see figure 4 (a)), shall be convoluted and shall be constructed of a minimum of 3 plies of nylon reinforcing fabric, in accordance with MIL-C-20696, impregnated with synthetic rubber in accordance with class 2, grade 60 of MIL-R-6855 and an outer ply of aluminized glass cloth in accordance with type 1, class 10 of MIL-C-20079. A minimum of 3 convolutions shall be used to provide for a total movement of 1 inch axially and 3/4-inch laterally. The expansion joint shall be further reinforced with round wire rings fabricated from CRES in accordance with QQ-W-423. The reinforcing rings shall not be bonded to either the cover or the tube of the expansion joint. Class B expansion joints shall have a 7 inch end-to-end dimension for all sizes. The inside diameter (i.d.) of class B expansion joints shall conform to outside diameter (o.d.) pipe dimension for all sizes (2 to 12 inch).

3.2.2.1 Retaining clamps. Retaining clamps for class B expansion joints shall be similar in design and construction to the clamp shown on figure 4 (b) and shall be fabricated of materials meeting the requirements of class 316 of QQ-S-763 and class 316 of QQ-S-766.

3.2.2.2 Physical properties, synthetic rubber (class B expansion joints). Physical properties of synthetic rubber used in class B expansion joints shall be in accordance with table V when tested in accordance with 4.5.2.

TABLE V. Physical properties of synthetic rubber for class B expansion joints.

	Initial	After aging
Tensile, lb/in^2 minimum	1500	1200
Elongation, percent minimum	300	180

3.2.3 Class C expansion joint. Class C, TFE fluorocarbon resin expansion joint shall be molded of resin conforming to class I, type III of ASTM D 1457. The expansion joint shall be a two-corrugation configuration with the ends molded outward over ductile iron flanges as shown on figure 5. The range of sizes and face-to-face dimensions shall be in accordance with table VI. The fluorocarbon resin shall have a minimum tensile of 2300 pounds per square inch (lb/in^2) and a minimum elongation of 400 percent when tested in accordance with 4.5.3.

TABLE VI. Face-to-face dimensions for class C expansion joints.

Pipe size inches ± 1/16	Face-to-face inches ± 1/16
1-1/2	1-3/8
2	1-9/16
3	2-1/4
4	2-5/8
6	2-3/4
8	4
10	5-1/4

3.2.3.1 Flanges for class C expansion joints. Flanges of the class C expansion joint shall be cast of nodular iron conforming to MIL-I-24137.

3.2.4 The manufacture of expansion joints shall not exclude the use of recovered material.

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3.3 Molding flash. Flash from molding process shall be carefully removed by trimming. Care shall be taken not to cut or gouge the body of the expansion joint when removing flash.

3.4 Marking. Expansion joints 12 inch i.d. and larger shall be marked with the following data, smaller expansion joints shall contain (a) through (d) only.

(a) MIL-E-15330, class and type, if applicable.
(b) Manufacturer's name or trademark.
(c) Serial number (class A expansion joints only).
(d) Date of curing (class A and B expansion joints only).
(e) Contract or order number.
(f) Nominal diameter.
(g) Flange face-to-face dimension (class A expansion joint only).
(h) The "H" dimension (class A expansion joints only, see figure 1).

3.4.1 Cure date. Rubber used in construction of classes A and B expansion joints shall have a cure date subsequent to the contract date.

3.5 Proof. When specified (see 6.2.1), expansion joints shall be hydrostatically tested in accordance with 4.6.4 to 150 percent of working pressure.

3.6 Vacuum. When specified (see 6.2.1), expansion joints shall be tested to 20 inches mercury vacuum when tested in accordance with 4.6.5.

3.7 Welding. Welding on reinforcing rings for class A, type 1 joints shall be performed in accordance with MIL-STD-278.

3.8 Fire retardance. Class A and B expansion joints shall be tested for fire retardance as specified in 4.6.6.

3.9 Radiography. Reinforcing ring welds shall be radiographed in accordance with MIL-STD-271 prior to placing the rings in the expansion joint body. This applies to rings for class A expansion joints only.

3.10 Drawings. The contractor shall prepare drawings in accordance with the data ordering document included in the contract or order (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. One complete sample of the largest expansion joint to be fabricated regardless of class and type, shall be submitted to the examination and tests of table VII, for those inspections requiring a complete expansion joint, to determine compliance to the requirements of this specification. Samples for those inspections which do not require a complete expansion joint shall be as specified in 4.4.

TABLE VII. First article inspection.

Inspection	Requirement paragraph	Inspection paragraph	Class		
			A	B	C
Visual examination 1/	---	4.5	X	X	X
Wire properties	3.2.1 and 3.2.2	4.6.1	X	X	-
Wrapping test	3.2.1	4.6.1.2	X	X	-
Rubber properties	3.2.1.6 and 3.2.2	4.6.2	X	X	-
TFE fluorocarbon resin	3.2.3	4.6.3	-	-	X

See footnote at end of table.

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TABLE VII. First article inspection. (continued)

Inspection	Requirement paragraph	Inspection paragraph	Class		
			A	B	C
Proof pressure 1/	3.5	4.6.4	X	X	X
Vacuum 1/	3.6	4.6.5	-	X	X
Fire retardance	3.8	4.6.6	X	X	-
Radiography 1/	3.9	4.6.7	X	-	-

1/ Tests requiring a complete expansion joint.

4.4 Quality conformance inspection.

4.4.1 Lot. For purposes of sampling, all expansion joints of the same class, type and size offered for delivery at one time shall be considered a lot.

4.4.2 Material samples. With each lot of expansion joints submitted for inspection, the following material samples shall be furnished:

(a) Six foot samples of each of the wire types used in the construction of class A and B expansion joints.
(b) Three samples, 6 inches by 6 inches by 1/16 inch, of each of the elastomeric composition used respectively for tube, friction stock and cover. Samples shall be thoroughly identified (class A and B expansion joints only).
(c) One sample TFE fluorocarbon resin for conducting tensile and elongation test of 4.6.3.1 (class C expansion joint only).

Elastomeric samples shall be identified as to material and state of cure with the material used in the manufacture of the expansion joints in the lot. If any sample fails to pass any examination or test, the entire lot which the sample represents shall be rejected.

4.4.3 Sample for fire retardance. A simulated expansion joint body section, containing all the materials used in the construction of a finished expansion joint, may be used for the fire retardance test. The simulated section shall be not less than 6 inches by 6 inches.

4.4.4 One expansion joint shall be selected from each lot for the tests in table VIII.

TABLE VIII. Quality conformance inspection.

Inspection	Requirement paragraph	Inspection paragraph	Class		
			A	B	C
Visual examination	---	4.5	X	X	X
Wire properties	3.2.1 and 3.2.2	4.6.1	X	X	-
Rubber properties	3.2.1.5 and 3.2.2	4.6.2	X	X	-
TFE fluorocarbon	3.2.3	4.6.3	-	-	X
Proof	3.5	4.6.4	X	X	X
Vacuum	3.6	4.6.5	-	X	X
Fire retardance	3.8	4.6.6	X	X	-
Radiography	3.9	4.6.7	X	-	-

4.5 Visual and dimensional examination. All expansion joints regardless of class and type shall be thoroughly visually and dimensionally examined. Each class A and B expansion joint shall be thoroughly examined after curing for evidence of voids, dimples, cracks, delamination, exposed fabric and cuts. The visual examination shall cover both interior and exterior surfaces to determine and evaluate defects in accordance with MIL-STD-407. The exterior fabric surface of the class B expansion joint shall be examined to assure proper adhesion to the expansion joint body. Shallow voids and dimples which do not expose the reinforcing fabric need not be repaired. Expansion joints shall be rejected for the following:

(a) Exposed fabric on inside surface of joint.
(b) Cut or damaged fabric anywhere on joint.

Exposed fabric on the external surface, except for class B expansion joints, may be repaired by vulcanization.

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4.5.1 Each class C expansion joint shall be examined for blisters, blow holes, cuts and gouges any of which shall be cause for rejection.

4.5.2 Marking examination. Any expansion joint which does not contain the necessary marking specified in 3.4 shall be rejected. Class A expansion joints shall be marked with raised letters with not less than 1/4 inch letter height. Class B expansion joints shall be stamped with a contrasting ink. Class C expansion joints shall be marked on the metal flange o.d.

4.6 Tests.

4.6.1 Wire.

4.6.1.1 Tensile strength. The tensile strength of the wire shall be determined in accordance with ASTM E 8. Wire used in class A expansion joints shall meet the requirements of 3.2.1 and wire used in class B expansion joints shall meet the requirements of 3.2.2.

4.6.1.2 Wrapping test. Each of the sample wires selected in accordance with 4.4.2 shall be wrapped six full tight turns around a mandrel of a diameter 3 times the diameter of the wire. The wrapped wire shall show no defects in the steel after wrapping. Flaking or peeling of the coppered coating of the wire shall not be considered a defect.

4.6.2 Rubber.

4.6.2.1 Tensile strength and elongation. Tensile strength and elongation of the synthetic rubber shall conform to 3.2.1.5 and shall be determined in accordance with method 4111 of FED-STD-601.

4.6.2.2 Aging test for synthetic rubber. Samples of synthetic rubber furnished as specified in 4.4.2 shall be subjected to dry circulating air at a temperature of 212°F ± 2°F for a period of 70 hours to determine conformance with 3.2.1.5.

4.6.3 TFE fluorocarbon resin.

4.6.3.1 Tensile strength and elongation. Specimens for determining tensile and elongation properties shall be prepared in accordance with ASTM D 1708. Method of testing shall be in accordance with ASTM D 638.

4.6.4 Proof pressure. Sample expansion joints selected in accordance with 4.4.2 shall be hydrostatically tested as follows with no evidence of failure or loss in pressure.

Class	Test pressure	Duration minutes
A	see table II	10
B	250 lb/in^2	20
C	300 lb/in^2	10

4.6.5 Vacuum test. Class B and C expansion joints shall be subjected to 20 inches mercury vacuum for a period of 5 minutes with no evidence of failure. There shall be no deformation of the internal reinforcing rings used on class B and C expansion joints.

4.6.6 Fire retardance. The fire retardance test apparatus shall consist basically of a Fisher-Todd No. 4-191 Universal Blast gas burner (or equal) mounted vertically about 18 inches under the test specimen. A thermocouple with an insulated lead shall be mounted about 1 inch below the test specimen in way of the flame. An insulated housing, with provision for visual observation of the test specimen, shall be placed around the burner, thermocouple and specimen supports. A controlled flame shall be applied to the test specimen. The flame intensity shall be increased until ignition of the test specimen is observed. The ignition temperature, as indicated by the thermocouple shall be not less than 1100°F. The flame intensity shall be further increased until the temperature indicated by the thermocouple is 1400°F. The burner flame shall be extinguished and the test specimen observed. The test specimen shall not support combustion as evidenced by cessation of all flame within 4 seconds after the burner flame is extinguished.

4.6.7 Radiography.

4.6.7.1 Reinforcing ring welds. Reinforcing ring welds shall be radiographed and examined in accordance with MIL-STD-271.

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4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. Expansion joints shall be individually unit protected in accordance with method III of MIL-P-116. Complete expansion joints weighing less than 90 pounds shall be unit packaged in a fiberboard box conforming to PPP-B-636 or PPP-B-640 with type, class or style at the contractors option. Box closure shall conform to the applicable container specification or appendix thereto, with method I and V closure applicable to domestic and weather resistant class containers respectively for PPP-B-636 boxes. Individual expansion joints 90 pounds or over shall be packed for the level specified in 5.2. For fiberboard boxes, load conversions shall be in accordance with the use criteria as specified in the appendix to PPP-B-636. Cushioning, dunnage and wrapping materials shall be as specified in 5.3.

5.1.2 Level C. Packaging of expansion joints shall afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractors normal retail or wholesale packaging methods may be utilized when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2.1).

5.2.1 Level A. Expansion joints packaged as specified in 5.1, shall be packed, except that expansion joints exceeding 90 pounds shall be individually packed in containers conforming to any one of the following specifications, with container selection at the option of the contractor.

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Overseas class 2
PPP-B-636	Weather resistant
PPP-B-640	Class 2
PPP-C-650	-------

5.2.1.1 Closure, case liners and gross weight. Shipping containers shall be closed, strapped or banded in accordance with the applicable container specification or appendix thereto, with method V closure applicable to PPP-B-636 boxes and reinforcing for fiberboard boxes shall be accomplished by use of nonmetallic banding or pressure sensitive reinforced tape at the contractors option. Unless otherwise specified (see 6.2.1), shipping containers shall have case liners conforming to MIL-L-10547. Barrier material conforming to MIL-B-13239 may be used to fabricate the case liner. Case liners shall be closed and sealed in accordance with MIL-L-10547. Open crates shall be waterproofed in accordance with the applicable crate specification. Case liners not required for PPP-B-636 fiberboard containers closed method V nor PPP-B-640 fiberboard containers when sealed with tape as specified in the appendix to the PPP-B-640 specification. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Wood or wood cleated plywood boxes shall not exceed 200 pounds except when the weight of an individual expansion joint exceeds 200 pounds. When the gross weight of a shipping container exceeds 200 pounds, the container shall be modified to include skids as specified in PPP-B-601 or PPP-B-621. Expansion joints packaged in weather-resistant fiberboard boxes as specified 5.1.1, closed and reinforced as specified herein will require no overpacking.

5.2.2 Level B. Expansion joints packaged as specified in 5.1 shall be packed in containers as specified in 5.2.1 except that the containers shall be of the domestic type or class and case liners are not required.

5.2.2.1 Closure and gross weight. Shipping containers shall be closed strapped or banded in accordance with the applicable container specification or appendix thereto with method I closure applicable to PPP-B-636 containers. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Wood or wood cleated boxes shall exceed 200 pounds except when the weight of an individual expansion

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joint exceeds 200 pounds. When the gross weight of a shipping container exceeds 200 pounds, the container shall be modified to include wood skids as specified in PPP-B-601 or PPP-B-621. Expansion joints packaged in fiberboard boxes will require no overpacking.

5.2.3 Level C. Expansion joints packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Cushioning, dunnage and wrapping materials.

5.3.1 Level A packaging, level A and B packing. Use of all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for material destined for shipboard use is prohibited.

5.3.2 Level C packaging and packing. Unless otherwise specified in the contract or order (see 6.2.1), use of all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for material destined for shipboard use is prohibited. When approved for use by the contract or order, unit packages and containers (interior and exterior) shall be marked or labeled as follows:

"CAUTION

Contents cushioned etc. with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material."

5.4 Marking. In addition to any special marking required (see 3.4 and 6.2.1), unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 and shall include cure date marking.

6. NOTES

6.1 Intended use. The expansion joints covered by this specification are intended for use in shipboard piping systems, where provision for expansion and vibration are necessary and where fire hazards must be eliminated.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Classes and types required (see 1.2).
(c) Type of drilling required for flanges (see 3.2.1.2).
(d) Dimensions and flange drilling (bolt circle/spacing, hole size and number) for class A, type II (see 3.2.1.2 and 3.2.1.5).
(e) Dimensional properties of replacement joints (see 3.2.1.5).
(f) When a specific arch construction is required (see 3.2.1.3).
(g) Hydrostatic proof required (see 3.5).
(h) Vacuum test required (see 3.6).
(i) When containers shall have caseliners conforming to MIL-L-10549 (see 5.2.1.1).
(j) When loose-fill materials are permitted (see 5.3.2).
(k) When special marking is required (see 5.4).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be

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selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Option
3.10	Drawing engineering and associated lists level 3 (production)	DI-E-7015	Design activity designation - Contractor Drawing number - Contractor Associated lists - Contractor Delivery of microfilm - Procuring activity

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4730-N579)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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